

Search sheets: how to run the activity



The pages after this one are 36 **search sheets**, dealt from *The Cat in the Hat* by Dr Seuss. Print them one-sided and hand out **one sheet per person**; every sheet is different, and nothing needs cutting out. Each holds 54–55 **token pairs**: a **next token** with the **previous token** that came before it in the text.

Match the room to the sheets, not the other way round. More people than sheets: print the set twice, so every count doubles and the proportions — which is all the model is — hold. Fewer: hand the spares out anyway, two or three to a person. A sheet nobody holds is a hole in the model.

In **in** **the**, the boxed token is the context and *the* is the next token. A token keeps its colour wherever it appears, so say the colour as you call one out — “who has *the*? it’s a purple one” — and the room can scan by colour before reading. With 8 colours and far more tokens, a colour narrows the search rather than settling it.

black **grey** **red** **brown** **green** **blue** **purple** **magenta**

Running a round

Seed the board: someone reads out any token pair from their sheet, boxed token first, and both go up — the room supplies the beginning too. Then, each round:

1. **Call out the last token** on the board.
2. **Hands up** from everyone holding a matching token pair — one hand each.
3. **Pick a raised hand at random** — whoever shouts first is the fastest reader, not a sample of the text — and ask for its next token.
4. The scribe writes that token up. It is part of the next context. Repeat.

What to watch for

The hands are the probability distribution. Every token pair sits with exactly one person, so a common continuation puts more hands in the air than a rare one. Nobody has to explain weighted sampling; the room performs it.

Nobody holds the model. No single sheet can continue the text on its own. One hand is not a stall — but no hands at all means an empty seat, so reseed from any token pair and say why.

1962 tokens, 240 unique · 2.83 bits/token · perplexity 7.1 · branching factor 4.32

Worked example

Start anywhere. Someone reads out a token pair — say this one — and both its tokens go on the board.

Round 1. Call out **the** — the last token on the board. A sheet across the room matches it, and that person reads out *milk*.

Round 2. Call out **milk** — the last token on the board. A sheet across the room matches it, and that person reads out *on*.

Keep going for as long as you like. What the room writes will sound like *The Cat in the Hat* without ever repeating it — that is the model generating, not the corpus being read out.

the token pair that answers

in **the**

the **milk**

milk **on**

on the board

in **the**

in **the** **milk**

in **the** **milk** **on**

in the game that fish in said ,
 things will Why do what would . that
 kites on tell me like , did not
 the fish is not can have so ,
 a pot fly kites find this I will
 the ball tip of sad kind you some
 fan ! us this , wet ! so
 have to do as should not ! look
 wet to had to saw them with the
 and I bump on . have I do
 he said we do that is milk on
 the way . and then I and so
 to you ! then not mind , I
 will show on a . not

find us did you Why , not all
! and and Thing day . and the
so tame , white said the hands with
a cake , what ! said is a
shame ! wet day kites , I call
! . . your with my then we
can hold put them I do . your
we had have Lots . he tip of
to us the milk will not saw him
you do the cup mess , on the
fish , like to he put fan as
do something house and ! said
that is the fish milk and him !
what to call up about it things and

then , him step mat ! fan ,
put me . you what will house .
? Oh you can call them on a
I said deep and wet , the cat
! and can hold to us . we
and Sally have any ! Thump hop up
with a things I ! Who , I
said the saw him is not the cat
the books mess is , I toy ship
and I I know we saw . and
he said like it hands . milk ,
not a will see . he about with
that are day . shame ! fish !
so so bad tricks do !

what to	him pick	! said	is not
the hat	. Yes	fish .	. two
house !	call this	with a	, with
are pink	said the	want to	toy man
so all	put them	shook their	hands with
you ?	will like	! look	fan with
bad ,	day .	wet and	like these
can hold	? Well	shame !	mess !
I will	! so	on there	he fell
we did	and so	do ?	about .
not know	that is	then Sally	. that
the hat	saw one	to find	and big
things fall	. do	hop on	, said
I will	the cake	deep in	

! what	things they	our fish	get them
gown with	fish in	then Sally	. you
wet day	house !	lit .	! and
the toy	toy ship	not sunny	to us
want to	day ?	I can	some good
we looked	he gave	on his	that bump
would you	what a	and Sally	. so
put me	so .	do not	I say
! now	good fun	will show	saw him
the cat	Think of	, said	a ball
can hold	said to	here .	like it
with a	the house	shook up	, no
you should	? Why	and we	are two
. .	there like	. Thing	

. then our fish ! and good ,
on a house ? and down . I
Sally . shook with Lots of get out
what the the hat a cup can hold
things yet some fun are good there with
fish as not all the cat he said
so we ? tell toy ship ! I
. But will not . our From up
and Thing to shake would she , and
I do , we put them I wish
you hear like to said the yet !
then I gown ! that . we saw
the hall kind of now you ! bump
here When not be at me

Sally and pick up that is ! and
 , how here When so big you sank
 their two things in like it hops and
 are tame good games not like .
 what did , and said the get those
 then those fish said ! sit they will
 house ! our toy not all Lots of
 to be the cat we saw .
 ? and a cup he came some new
 . I on the and Thing ! he
 . and I like there , would you
 the cat at all gown , now look
 bit ! of her can hold I said
 and this will she the hat

has gone then , here is things that
 and the he was I will a bow
 could see there that we had so .
 will hit be in at this not like
 I will . what . my ! then
 can get that is ! we they will
 their kites bit ! on the our mother
 said the not be ! you some milk
 are not fish said pick up and the
 ? I like that good tricks what to
 now ! you wish . I , wet
 the fish of the get rid the cat
 house ! to get , too bite you
 . we Sally and the cake

tail I that is good trick he lit
gave them things . like to dish !
of this be here cold to can fan
! then and the PLOP ! now !
what you on my not . , not
here to they play we saw . .
? what at all ! said then our
not shine house ! a pat hear !
our mother to fall there on you up
could , the rake the cat said the
fish . pick it I can I will
. he so we . But , said
. and Sally and will have For the
bit . ! those the fan

out he got to do into a
hear ? ! the a look , then
! and and I our fish fish shook
! what were down pick up dish ,
things . I will not in could do
Sally and there is the hat be here
here . at me on the the fish
they like ? should in the will show
how do fast as now ! house all
! she that were of this .
cold , I do not fly the hook
said , pack up can get ! Oh
like this what will good game .
we can then he you pack

we saw

a rake

fast as

of good

in the

can hop

will not

. he

be here

kicks and

not be

books !

up and

! I

on the

man !

and all

, she

things ran

I can

could play

our fish

kinds of

give you

, what

that I

cold ,

. we

fish .

you sit

! I

cat go

. and

string of

house For

how that

what would

I hop

the cat

the dots

? Oh

good things

said to

to know

then something

now ,

here When

he shut

. .

at that

Sally and

like the

the house

! look

they like

said to

can have

. these

be here

, we

things they

? now

one and

in with

two things

the day

what a

and the

not want

how I

on the

at me

will give

strings ,

to the

a book

then he

! so

and Thing

! if

box ,

I ,

now look

playthings and

books ,

like it

up these

of something

the pot

mother's new

! they

. then

. and

our house

fish in

. I

we saw

net .

wood box

fast .

, so

you like

that I

lot of

I can

show you

cat in

the ball

they are

good fun

I call show to I can box are
like our one little the house our new
a tip , said ! said to say
you another trick that little toy , the
! her now , net . in the
! put out of , Why . this
cat in lot ! fish said what a
of good . it . and they should
pat . two books . Oh good game
things that said the not be we sat
how . will not the box play .
books ! at all be here and kicks
then he fast as that day and so
? would up the the fish

, with . so . I to the

the things then he trick , fast !

out and we can net , last !

will come looked ! they should the hat

Thing one and a Oh , two things

! do I can fish came pot !

tricks . one and the fish cat in

said the . a in the , do

a box our mother and look now .

at last ! and be about box .

I do you have ! said that is

up on , said show them not one

came down play ! no fear little fish

. this ? Oh of look

no , . I I said show you
up those box with you do we sat
. have pot . will bump the cake
then our the mat two , Oh ,
that they out ! little bit not wish
Thing two . the ship and red
looked ! now look one one ! at
, no said the of his in the
, and and I net I funny !
sat in , here they will cat .
came in our fish at me ! those
hall . ! put sun did a bump
fish . play , tricks . to go
. they I can ? and

of Thing Oh dear and the . there
our fish box . in the . I
little bit ! how ! then two .
them go ! then not . one and
. with cat in said , play ball
pot . , what tricks , no fear
Thing two bow . and you hall .
out ! take them will find the cat
we sat this mess now ! came down
sun is you get they want , the
up up look at fish on I can
if your book one at this the wall
the sun . that I let a lot
, I to do sat there

, jumps not pick . you our game
 look ! if you to do and the
 in a and a take a fall .
 When your of one , how do not
 we looked they ran play . ! said
 the hook the cat little bit two things
 fear . . I the milk at all
 this mess said the came down , look
 a big them a no fun I do
 out of ! have home ! I saw
 box came ! I Oh , one hand
 up high pot . run down sat there
 Thing one his hat cat . tricks are
 you will fish to . it

up . at me ! they cat .

When our in the the cat came Thing

, now of the hand ! the cat

and Thing . and with Thing

to the , I sat in no fear

you now we tell Oh , the wall

those things I call new tricks play .

do ? they have look at asked you

one ! his hat I always said the

fear ! them out , said . no

out ! fall ! Thing one if I

not know ! then little toy a little

pot . ! said bed ! two things

and I this game

Thing one new . at this a tip
came back this way Well . we saw
the house , and out of not be
. they When your said , cat .
went on and Thing his head , said
and they I had fall ! look at
no fear ! I fear ! ! now
. I . in is what ! he
Make that one ! up all the cat
with a you do of my do .
Oh , tame . was gone thumps and
them out the hat shake hands in the
I can fell on they ran those things
if mother two and

cat with said the out . a fish
! put . . said our tame !
those things ! and I bet went away
cake ! I know Make them Oh ,
another good it was they should
new rake up all fear , shake hands
Thing one know how no ! , that
the fish and the fell into . I
wall ! with big not know do not
 . you look at this house and you
the things ! and of his in the
 , said them I one little two !
is not the fish was out mind at
you did say . we could

! they , said sank it Oh ,
my tricks cat , Thing two .
the things was to say . and our
they said two ? the hat . and
the pot , and up the a box
with hops is not fear of sit !
said the of good I stand went bump
look ! bet . ! have ! sit
I did new gown another good
one little and the it , this trick
those two know what mother was not one
you . all the down on do not
them Thing wall in in the cake .
said the . I no fear

one kite

no !

do .

as I

Oh ,

they ran

all .

me !

I sat

my tail

! they

look what

them to

the cat

sank our

not let

! should

fear .

said the

! bump

the cat

come here

, as

and he

is not

, sank

! it

went after

was too

her the

two things

and so

up there

wish we

with a

say .

sit !

! your

cake ,

know !

I like

cat .

of bad

those two

it deep

bet ,

in the

Thing two

said ,

this wet

! not

down the

mother could

a red

was shut . so then come .
those things ! what sit ! say ?
step in ! and is good up our
cat went wish to and some I know
the string down with me . them away
bumps , one kite I went , no
a dish two . do you as I
cake on mother is look ! with a
to have , and ! your the hat
big red not ! hat . this box
all kinds fun to and our . now
it a my playthings no !
her . said , Thing two know it
Oh , way that in this

the fish the cat to play cat !
 and look . and not bad and he
 said , in the away . it was
 a sad all we ! and them away
 sit there way ! high as my hat
 her way mother is , if , said
 ! down Oh , you will do I
 is near fun in no way as fast
 I could was back Who was gone with
 down my up up say ! one and
 me now ! said . you hat !
 two . this is big and with a
 Thing one know some . I wish ,
 pink , look at

picked up

, we

to us

, wet

mother !

is wet

. look

as a

you take

gone away

no !

up !

! look

her about

look at

this !

them run

two and

the house

and the

fun on

a house

I did

. Oh

down !

hat you

Oh ,

away !

cup on

know !

! said

! Make

the cat

ship ,

Thing two

me !

said .

my net

But that

these things

wish to

way home

it came

all !

and he

. with

nothing at

in and

cat .

sit !

not bite

do you

big bumps

with a

mother will ! fast a game all .
, and up with tall , stand on
fun at ! said hat . know what
do not in the way at them !
big thumps said the not like no .
two hands cat in and the the hall
I will dear . . Oh . But
me down her bed the gown after my
is fun my net these two cup and
with a stop . . a it is
Oh dear this ? you hear , no
But you and the away . look !
Yes . down the to play ! But
ship ! as I

look it! I like down!
 do was is that dear! fun?
 no away! and then have
 that cat something went my net
 a good , said . hat.
 ball! her gown ship, said the
 But your look! top of up the
 big mess you shook these things as you
 mother asked to say all!
 have to . not one the top
 cat. me down know some ! we
 let you should we with the cup,
 the cat and I , did back in
 in a rake. this is no!

this house should not a game fun But
 it is , said have no ship and
 my net hat . ! and me !
 no . down . let down cat in
 ! said the fish she will ! I
 know what look at all ! and I
 . and away with ! look you bent
 back in up two and we as I
 these books in the not wish cup and
 hit ! I did But I . then
 said the is not mother will do you
 the cat top with , we rake ,
 with a ball . to stop . I
 that they something to

in the ! and all . ! then
 ball . ran up to play a shame
 my hat and the hat to , as
 ! and is so did not rake !
 kite bump this , you . she do
 hook . look at that is down ,
 . we up the fun that and Sally
 your mother me now should not sunny .
 cat . with the as he no !
 said the the things have no I can
 . . it to But that do ?
 something new ! too go !
 . they not all the sun any fun
 mother came away . , with

so, the cat go away . now
something to should not do !
! take and she , Oh hold up
the box in on she say hat ,
kite we cat in ball . up the
Thump ! have fun But our hook .
! I ! sit with my I saw
got up it . mother is bump !
ran down a PLOP said the is on
did nothing to fly down with .
! look ! I fun , look .
not . your mother , and that I
you do this house as fast he should
too . all the and a

But little have no and
 bump made it and I is out
 fun . in the to you do not
 all . he picked ! on with a
 should not that ! so deep did you
 I know too wet it ? we can
 they she will cat . ! then
 ! put mother is your mother games we
 you do not like and a hook .
 look at a little something ! ran to
 Thump ! the things always pick head !
 , and go away said the ball !
 down ! But that this big hat !
 the cat hold you

and then , cold not like ! no
 things had said the it up you fall
 will hold all I jump ! cat in
 ran about he should game that the pot
 on top near ! to sit with Sally
 ! Oh hold up should we she say
 ! you bump ! your mother . Oh
 did not go out that is mother is
 and with . he . he hat .
 we shook shut with . now so so
 too cold head of in the , and
 then we ! and have Lots do to
 us two I said But we a shame
 is no the ball

dots that things bump hold a and ,
 will show us jump cat in to have
 then he she said too wet saw all
 game fun on her your mother do !
 said the the cat ! they have no
 a fox ! tell bump From you have
 should we with my , fast . you
 we two red . hat . he should
 , no is no . you they
 I bet . But go ! ! But
 and the the fish in the jumps and
 that ? shut up But our did not
 not like the things so we ran out
 all that ! with

will pick should not your mother us ?
tell her do you I do to shake
bump ! on the with fear you another
then ! the fan in this that cold
game . red wood ! Make , I
fox , fish in and too cat .
have fun not be had something
a hook hold up said the , Think
rid of is not we did ! Oh
made us . that ! said see this
. I he fell so tall ! Thump
saw those the fish things should
can not hat ! . then all if
did ! the cat and then . Why

not go white and the hat that cat
your mother ! and cat in I know
fish . and the then he will she
so do all my he ran ! they
did not on this said , to play
tell her , the the strings bent our
red fan should not ! I can ,
us fast . then . Oh and I
hold up the pot wet to . then
a shame hat came game , bump their
you did had them do if have no
. I saw mother's with a
things are see something we could
is funny in the ! Oh , said

so , have come on the . these
 , so the head and I I can
 . what do not and then hold the
 cat said kites in he has ! But
 bump ! we tell should not us ,
 not all Why , what should saw her
 tell that things , ! not to go
 fly kites ! now game that fish !
 , the did not that went can do
 . no then we said , shine .
 with Thing the dish like it in the
 will not a lot the cup . so
 you will is not your mother had to
 hat . wet ,